



# THE TRANSPORTATION- SOCIAL EQUITY TOOLKIT

Data to prioritize disadvantaged communities in transportation planning



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# I. The Equity Imperative

Transportation is now widely understood as one of the cornerstones of social equity. It's the means by which people access jobs, healthcare, and meet day-to-day needs. But social equity has not always been a consideration in transportation planning. Much existing transportation infrastructure — such as intra-city urban highways, limited transit service, and dangerous road design — is now seen as unequally harming certain communities. As a result, projects like highway caps, increased transit service, safety planning, and vehicle electrification infrastructure are gaining attention as a means of improving social equity among disadvantaged communities, in addition to fostering broader public health, economic, and climate benefits. Equity is also being considered as an input for prioritization of more basic project work, such as roadway maintenance, repaving, and congestion mitigation.

As more attention is paid to the relationship between transportation and social equity, agencies and transportation planning firms are placing increased emphasis on measuring the social equity impact of transportation and advancing projects that aim to redress historic harms to disadvantaged or underserved communities.

In addition, some grants give extra weight to applications that show how a given project will benefit underserved communities. The federal government has even implemented the Justice 40 initiative, which specifically defines disadvantaged communities and targets them for increased funding.

**But that still leaves open the question of how to analyze social equity and prove impact.** A single definition of equity does not exist, plus governments have limited resources and inequity looks different in every locality.

To better diagnose and remedy social equity in a transportation context, it's critical to analyze:

- Travel behaviors by demographic group to understand transportation patterns affecting underserved constituents
- Travel behaviors occurring within disadvantaged vs. non-disadvantaged communities to measure disproportionate impact

## II. Defining & Prioritizing Equity in Planning

Understanding travel behavior by demographic groups, including rural vs. urban, income, education, racial makeups, family profiles, languages spoken, and more helps planners see how underserved communities travel and what infrastructure can better serve them. In addition, mapping transportation activity within disadvantaged communities, such as those defined by the Justice 40 initiative, is useful for analyzing how travel impacts underserved communities where they live.

Here are some questions a transportation leader might ask to identify opportunities for social equity investment:

- Does the burden of commute time on lower-income families justify a new bus line?
- Which disadvantaged rural communities would benefit most from EV commercial chargers?
- Should we add multilingual signage for non-English speakers?
- Should the team revise its maintenance schedule to improve conditions where underserved individuals travel?

Looking across a city, county, or state, the need to reduce congestion, improve air quality, shorten commute times, provide multimodal infrastructure, and offer sufficient EV charging stations is enormous. To assist transportation leaders in making critical prioritization decisions and clarifying those priorities to the public and stakeholders, analyzing disparities in how different demographic groups travel and how travel occurs within disadvantaged communities is essential.

[StreetLight's Traveler Attribute Metrics](#), derived from U.S. and Canada Census data, include a wide array of demographics most recently updated to incorporate the 2020 U.S. Census and 2019 5-year American Community Survey.

In the StreetLight InSight® platform, you can review trends through a wide spectrum of census demographic breakouts, including:

RACE / ETHNICITY

EDUCATION

EMPLOYMENT CHARACTERISTICS

EMPLOYMENT INDUSTRY

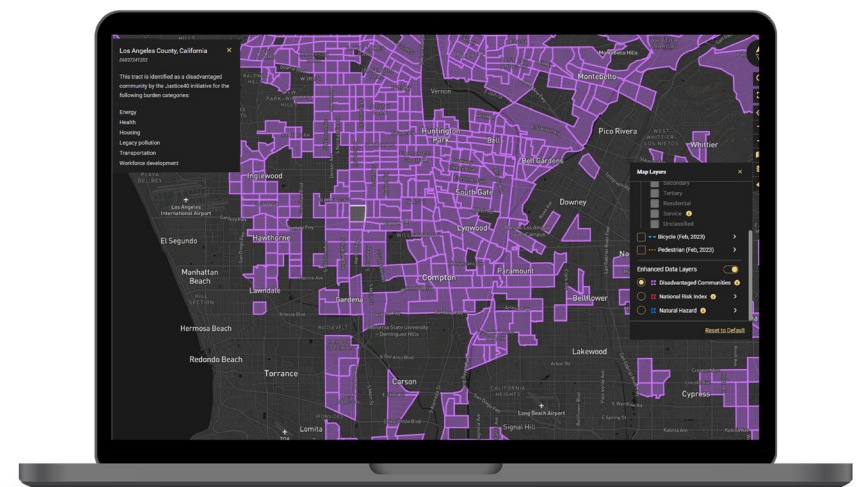
HOUSEHOLD INCOME

VEHICLE OWNERSHIP

FAMILY STATUS

HOUSING TYPE

Disadvantaged Communities (DAC) layers in StreetLight InSight® enable planners to understand any travel behavior happening within these burdened communities and compare against surrounding non-burdened communities from their desktop.



# III. Equity Planning in Action

## Capping a Nashville Highway — What Traveler Demographics Reveal

I-40 in Nashville, TN is known for having bifurcated a low-income minority community, creating barriers to downtown Nashville and exacerbating public health and economic challenges. While I-40 has a number of bridges crossing the highway, some have narrow sidewalks and wide travel lanes, creating unsafe conditions for vulnerable road users trying to cross the bridge without a vehicle. In addition, bridges limit easy connectivity between the east and west sides of the highway compared to a highway cap, which would create a wider deck over the interstate.

Recognizing this historical inequity, Nashville's 2020 transportation plan<sup>1</sup> calls for putting a cap over the highway at Jefferson St. where there is an existing bridge. The cap could expand multimodal options, increase green space, reconnect communities, and benefit those on both sides of the highway economically.

To understand whether capping the highway at Jefferson St., would indeed support social equity, StreetLight used traveler demographics to analyze the travelers at this juncture along the highway as compared to other nearby bridges across the highway. The data reflects all vehicle travel between January through March 2024.

### STREETLIGHT'S FINDINGS REVEAL:



Users of Jefferson St. bridge were 51% non-white or multiracial as compared to the composite demographics of other I-40 bridges, at >62% white



36% of Jefferson St. travelers had a High School Degree or less, compared to 32% on other bridges



43% of Jefferson St. travelers had a household income of \$50,000 or less, compared to 35% on other bridges

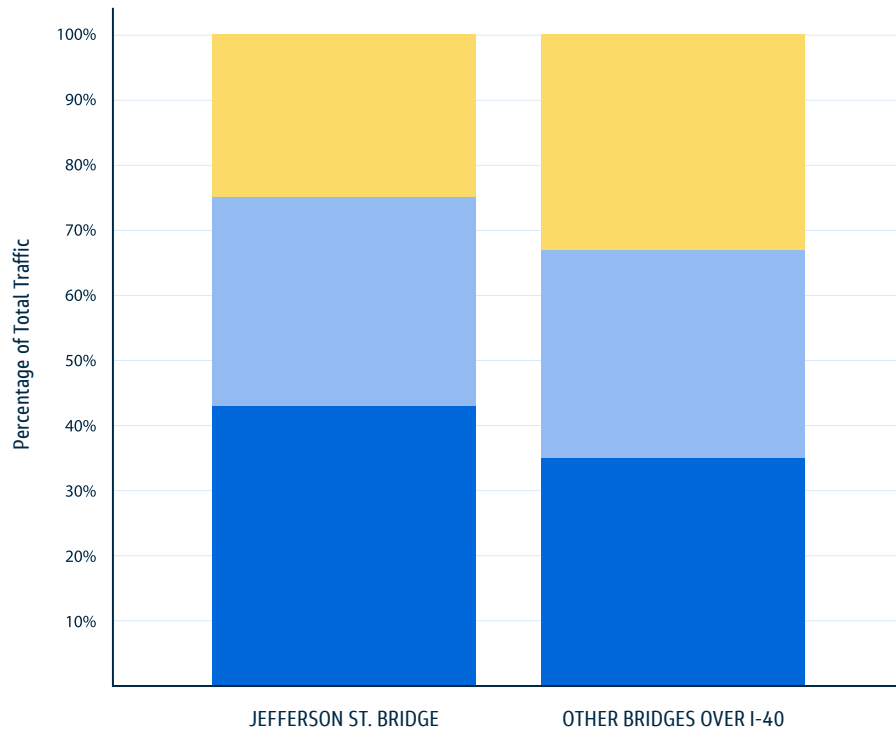


1. Metro Nashville Transportation Plan, Office of Mayor John Cooper. <https://www.nashville.gov/sites/default/files/2022-04/Metro-Nashville-Transportation-Plan-2020.pdf>

## TRAFFIC ACROSS I-40 BRIDGES BY INCOME

Distribution Of Traffic By Income, Jan.-Mar. '24

- Income <\$50K
- Income \$50K-\$100K
- Income >\$100K



The data shows that historically disadvantaged demographic groups did use the Jefferson St. bridge at a higher rate than other bridges across I-40, indicating that a cap here would benefit these historically underserved travelers.

StreetLight's analysis also finds that the bridge could provide additional benefits based on typical driving behaviors by potentially providing safe and comfortable conditions to walk or bike, rather than drive. This carries the add-on benefits of potentially helping reduce emissions and lessen congestion. Specifically, **more than 27% of vehicular travel over Jefferson Street were trips of less than 5 miles (compared to about 21% of trips under 5 miles across nearby bridges), and nearly 7% were less than 2 miles.** Some share of these nearly 5,000 average daily trips could potentially be served via walking or biking.

# How Port of Norfolk Freight Activity Impacts Disadvantaged Justice 40 Communities

Truck activity is on the rise. While freight is critical for economic output, increased diesel and gas-powered truck activity can negatively impact air quality, congestion, noise pollution, and safety. This issue is [particularly salient among disadvantaged communities](#), which are often located near port and warehouse hubs, and thus may be disproportionately affected by negative outcomes as truck traffic increases. This inequity may also be exacerbated by where truck demand is generated. Trucks often travel through disadvantaged communities to service demand for goods from farther flung and wealthier locales.

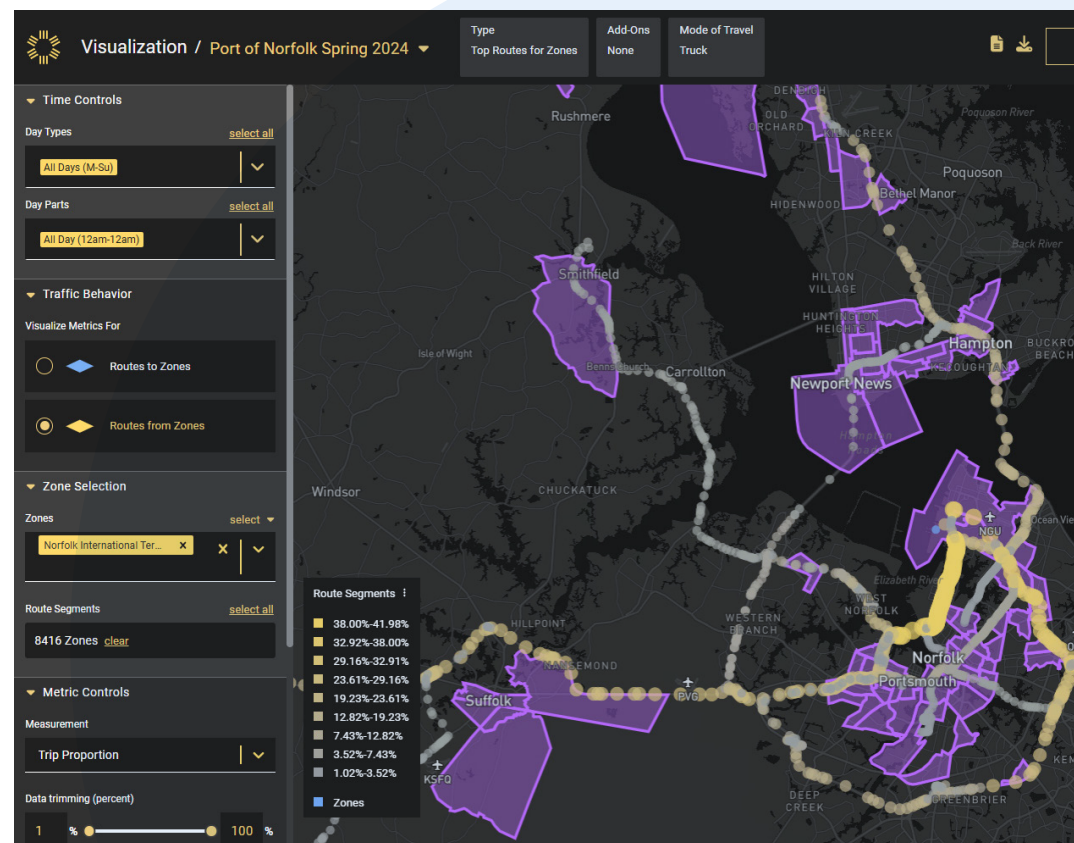
To better understand how truck traffic disproportionately affects disadvantaged communities, StreetLight analyzes the top routes of trucks leaving the Port of Norfolk in Virginia, one of the biggest and fastest-growing ports in the country, and visualizes these routes mapped over Disadvantaged Community layers. This provides an easy-to-read rendering of which Justice 40 communities are most commonly affected by truck activity coming out of the ports. It also highlights whether truck activity most commonly ends in these communities to support local demands, or passes through.

StreetLight's analysis shows major truck activity originating from the Port of Norfolk was routing through DACs concentrated around the ports, as well as in DACs over 100 miles away in places like Charlottesville, VA; Rocky Mount, NC; and Washington, DC.

By pinpointing affected communities, agencies, planners, and firms can start to identify targeted mitigation opportunities, including routing truck traffic away from local roads, commercial electrification priorities, and more efficient routing.

Following a top routes analysis, planners can:

- Use congestion metrics to understand critical bottlenecks to reduce congestion and emissions in DACs.
- Use Origin-Destination and trip attribute metrics (e.g., trip length, average speed) to identify popular trips that could be good candidates for alternative fuels.
- Dig deeper into Top Routes to identify unnecessary freight cut-through traffic, or deviations from preferred freight routes.



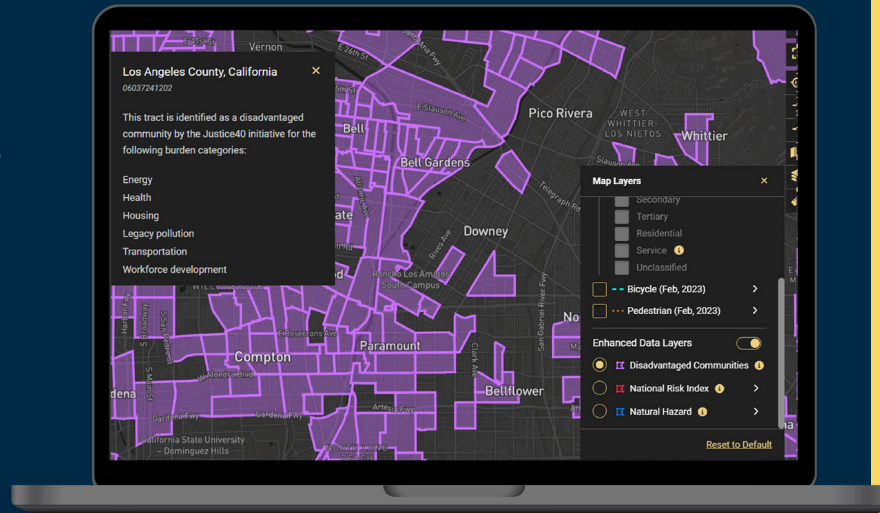
StreetLight InSight's Top Routes shows major truck activity originating from the Port of Norfolk. Purple areas show DAC communities.

Regions that have defined disadvantaged communities or communities of concern according to different criteria than Justice 40 can use top routes to identify similar impacts in priority areas.

# Ready to measure social equity in your transportation planning?

Use StreetLight InSight® to zero in on your top opportunities to improve equity in transportation.

- Measure travel behaviors at a high resolution and analyze activity by demographic category.
- View travel behaviors through Justice 40 disadvantaged communities.
- Understand travel patterns to prioritize safety and connectivity for Vulnerable Road Users.
- Use data and visualizations to show the value of social equity prioritization decisions.



## What is StreetLight InSight®?

It's self-serve software that lets users ask mobility questions and get answers within minutes. Access traffic metrics for any road, without sensors. From vehicular O-D and VMT to bicycle, pedestrian, and commercial truck metrics, **point and click** your way to the data you need to measure climate policy success.

Learn more at:

[streetlightdata.com/  
transportation-metrics](https://streetlightdata.com/transportation-metrics)



[streetlightdata.com/transportation-social-equity-solutions](https://streetlightdata.com/transportation-social-equity-solutions)

# About StreetLight Data

StreetLight Data, Inc. ("StreetLight") pioneered the use of Big Data analytics to shed light on how people, goods, and services move, empowering smarter, data-driven transportation decisions. The company applies proprietary machine-learning algorithms and its vast data processing resources to measure travel patterns of vehicles, bicycles, and pedestrians, accessible as analytics on the StreetLight InSight® SaaS platform. Acquired by Jacobs as a wholly owned subsidiary in February 2022, StreetLight provides innovative digital solutions to help communities reduce congestion, improve safe and equitable transportation, and maximize the positive impact of infrastructure investment.

